

Replayable Minerva And Athena Ballot-Polling RLA Transcripts

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Abstract

Minerva is a round-based ballot-polling risk-limiting audit: it groups sampled ballots into audit rounds, recomputes a binomial-tail likelihood ratio from cumulative observations, and stops when the round statistic reaches the risk limit threshold. Athena is the related public-method surface that RCOUNT now records without claiming risk replay. This paper documents the RCOUNT Minerva/Athena contract. A verifier reads the method id, reported two-candidate totals, risk limit, optional round boundaries, and ballot observations from an audit record; routes Minerva records to the exact round replay in `rcount-stats`; and emits an explicit boundary transcript for Athena or for Minerva records missing required inputs. The claim is one layer below the statistical policy question: when sufficient public evidence exists, the declared round transcript is reproducible from the declared record.

1 Introduction

Ballot-polling RLAs are often described by their stopping rule, but the public record also has to support replay. BRAVO, documented in V.12, is per-draw: each sampled ballot updates a sequential likelihood ratio. Minerva moves the same ballot-polling idea to audit rounds. The replay question is therefore not only whether a final stop was declared, but whether each declared round boundary and round statistic follows from the reported totals, risk limit, and observed ballots.

RCOUNT treats this as a transcript contract. The verifier does not accept a vendor-exported risk value as authoritative. It partitions sample steps into rounds, recomputes the Minerva binomial-tail ratio with exact rational arithmetic in `rcount-stats`, and compares the computed round statistic, p-value in parts per million, and stop decision to the declared audit record.

This paper covers that Minerva slice and the related Athena method surface. It does not claim that Rhode Island Rep. 28 or any other public audit has enough released ballot evidence to recompute its Minerva risk value. It claims that synthetic two-candidate Minerva records with sufficient observations are replayable, and that Athena records are preserved as method-specific boundaries rather than silently treated as generic unsupported methods.

2 Minerva Contract

The audit record carries the public inputs the implemented Minerva replay consumes:

Field	Meaning
<code>method_id</code>	Versioned method name: <code>minerva-ballot-polling-v1</code> or <code>athena-ballot-polling-v1</code> .
<code>risk_limit_ppm</code>	Risk limit in parts per million, used to form the stopping threshold.
<code>reported_winner_votes</code>	Reported votes for the winner of the two-candidate assertion.
<code>reported_loser_votes</code>	Reported votes for the loser of the assertion.
<code>sample_steps</code>	Ballot observations. Assorter value 1/1 is a winner ballot and 0/1 is a loser ballot.
<code>round_index</code>	Optional zero-based round marker on each sample step. If any step has it, every step must have it and the sequence must be nondecreasing.
<code>statistic, p_value_ppm, decision</code>	Declared round outputs checked against the replay.

When no `round_index` values are present, the run is treated as one round and the final sample step is the declared round step. When explicit rounds are present, the final sample step in each round carries the declared round statistic. For each round, RCOUNT adds that round’s observations to the cumulative sample, counts winner ballots, computes the binomial upper tail under the reported winner share and under the null share 1/2, divides those tails to obtain the likelihood ratio, converts the inverse statistic to a conservative parts-per-million p-value, and stops when the ratio reaches $1/\alpha$ for the declared risk limit α .

3 Implemented Surface

The algorithm dispatcher in `rcount-audit` routes `minerva-ballot-polling-v1` to `replay_minerva_run`. That function requires `risk_limit_ppm`, `reported_winner_votes`, `reported_loser_votes`, and at least one sample step before it calls `replay_minerva_ballot_polling_rounds` in `rcount-stats`. The statistics crate owns the cumulative round replay and returns one computed round step per declared round.

The same dispatcher routes `athena-ballot-polling-v1` to a documented boundary transcript: “Athena round risk calculation is recorded but not replayed.” Minerva records also become boundaries, rather than failures or passes, when required fields are missing, explicit round indices are incomplete or decreasing, a round is empty, reported vote totals are invalid, or a sample step contains an observation outside the implemented winner/loser surface.

Check	Failure caught
<code>audit_algorithm_run</code>	Unsupported method id, Athena replay request, or Minerva record missing risk limit, reported winner votes, reported loser votes, or sample steps.
<code>minerva_rounds_from_steps</code>	Partial or decreasing explicit <code>round_index</code> values, and sample steps that cannot be reduced to winner or loser observations.
<code>minerva_replay</code>	Declared round likelihood ratio, p-value, or stop decision drifts from the recomputed cumulative round transcript.
<code>cvr_summary_total</code>	Reported contest totals still reconcile to the public summaries that anchor the assertion.
<code>source_hash_match</code>	Source evidence bytes still match the declared source index.

The result is deliberately narrow. A passing Minerva transcript means that the declared two-candidate round transcript follows from the record RCOUNT was given. It does not expand Athena into Minerva, and it does not infer missing round evidence from a public report.

4 Fixtures

The round-one positive fixture is `synthetic_minerva_round_one_package`. It starts from the synthetic summary package, adds one `minerva-ballot-polling-v1` run for `syn-2024-mayor`, reports winner and loser votes as three and one, sets a ten percent risk limit, and supplies six winner observations with no explicit round index. The test `minerva_round_one_algorithm_statistics_replays` checks that the replay emits one round step on sample step 5, computes the fixture statistic, and stops.

The multi-round fixture is `synthetic_minerva_multi_round_package`. It reuses the same sample steps but marks steps 0 through 4 as round 0 and step 5 as round 1. The test `minerva_multi_round_algorithm_statistics_replays_round_boundaries` checks that round 0 continues and round 1 stops, proving that the statistics crate uses cumulative observations across explicit round boundaries.

The Athena boundary fixture is `synthetic_athena_boundary_package`. It changes the multi-round run's method id to `athena-ballot-polling-v1`, sets the decision to `Boundary`, and verifies through `athena_algorithm_statistics_reports_documented_boundary` that the method-specific boundary text is preserved.

The drift case is covered by the CLI test `replay_audit_algorithms_exits_one_on_minerva_declared_drift`. It edits the declared `p_value_ppm` on the final round-one sample step while leaving the observations fixed. The replay exits nonzero and reports a declared p-value mismatch, separating a reproducible non-stop from a declared transcript that no longer matches its public inputs.

5 Boundaries

This slice verifies a ballot-polling outcome assertion under the sampling and round-planning assumptions already encoded in the audit record. It checks that a declared Minerva stop-or-continue transcript follows from reported two-candidate totals, a risk limit, round boundaries,

and winner/loser ballot observations. It does not verify ballot custody, voter eligibility, manifest construction, public-seed sampling, or ballot-level CVR comparison; those are separate RCOUNT contracts.

The method boundary is equally important. The implemented Minerva path is the scoped two-candidate ballot-polling replay in `rcount-stats`. Athena is currently a record-shape and method-preservation boundary only. RCOUNT may say that an Athena audit record was carried through the package and identified by method id, but it must not claim Athena risk replay until the required method-specific inputs and tolerance rules are implemented.

The public-evidence boundary remains with the artifact. The Rhode Island Rep. 28 adapter records MINERVA metadata and emits a package-level Minerva run, but the plan explicitly keeps that source as boundary-only until public rows expose enough per-round ballot evidence to replay the source risk measurement. Missing evidence is therefore reported as a boundary, not upgraded to a trusted pass.

6 Conclusion

RCOUNT now has a replayable Minerva ballot-polling contract for synthetic two-candidate records with sufficient ballot observations. The verifier reads reported winner and loser totals, the risk limit, optional round indices, and sample-step observations; recomputes each cumulative round statistic in `rcount-stats`; and compares the declared likelihood ratio, p-value, and stop decision against the recomputed transcript. Missing Minerva fields and all Athena records produce explicit method-specific boundaries.

This extends the BRAVO paper by moving from a per-draw sequential transcript to a round-based transcript. The statistical machinery differs, but the public contract is the same: a named audit method should be either replayed from its declared evidence or bounded as unreplayable, never silently trusted.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Replayable bravo ballot-polling audit transcripts. Technical report, Apportionment Research, 2026b.